

Machine Learning Practical Documentation

From Theory to Implementation: Data Preprocessing & Feature Selection

Course: Machine Learning	Project Type: End-to-End Practical Implementation
Domain: Industrial Fabric Quality Inspection	Target Variable: fabric_quality (High, Medium, Low)
Philosophy: Theory → Formula → Scratch Code → Library Verification → Interpretation	GitHub Repository: YashCube-x/Alok_sir_ML-project

1. Project Purpose & Problem Understanding

What is this process?

In textile and apparel manufacturing, fabric quality directly determines product reliability, consumer safety, and commercial value. This project implements a complete, leakage-free Machine Learning preprocessing and feature selection pipeline from scratch to automate the classification of fabric rolls into **High**, **Medium**, and **Low** quality grades based on sensory, physical, and process parameters.

Why use Machine Learning here? (Real-World Impact)

- Manual Inspection Limitations:** Human visual inspection is slow, subjective, prone to fatigue, and misses micro-level structural inconsistencies (e.g., tensile stress variances or non-uniform moisture absorption).

- **Automated ML Advantage:** Standardizes inspection across batches, detects physical defects in real-time, reduces fabric wastage by over 25%, and prevents defective garments from reaching consumers.

2. Exploratory Data Analysis (EDA) & Descriptive Statistics

What is this process?

EDA is the foundational step where we mathematically profile each attribute to understand its

central tendency (Mean, Median, Mode), spread/dispersion (Variance, Standard Deviation,

Range), and distribution shape before applying any transformations.

Mathematical Formulas:

- Mean: $\bar{X} = (\sum X_i) / n$

- Sample Variance: $s^2 = [\sum (X_i - \bar{X})^2] / (n - 1)$

- Sample Standard Deviation: $s = \sqrt{s^2}$

- Range: $X_{\max} - X_{\min}$

Why do we use it? (Example):

Consider `tensile_strength`. Calculating Mean (67.52 N/m) and Standard Deviation

(30.30 N/m) tells us the expected baseline strength of fabric rolls. Comparing the Mean to the

Median reveals whether the feature is symmetric or skewed by anomaly recordings.

3. Missing Value Handling & Imputation Strategy

What is this process?

Detecting unrecorded, corrupt, or missing observations in numerical and categorical features and

replacing them using statistically sound estimation techniques rather than deleting records blindly.

Missing Percentage Formula:

$$\text{Missing \%} = (\text{Number of Missing Values} / \text{Total Number of Observations}) \times 100$$

Data Type	Imputation Method	Why This Method Was Chosen?	Real-World Example
Numerical Features (e.g., tensile_strength, gsm)	Median Imputation	Mean is heavily influenced and distorted by extreme outlier values. Median represents the 50th percentile rank and remains completely robust against extreme values.	If gsm has values [150, 160, 165, 1150 (outlier)], the Mean is 406.25 (distorted), while the Median is 162.5 (accurate reflection of typical fabric).
Categorical Features (e.g., fabric_type, weave_type)	Mode Imputation	Calculates the most frequently occurring categorical value in the training distribution.	If 70% of inspected rolls are cotton, an unrecorded fabric type entry is statistically most likely to be cotton.

4. Duplicate & Invalid Data Treatment

What is this process?

Data cleaning involves detecting exact duplicate rows, physically impossible negative values,

whitespace artifacts, and casing inconsistencies resulting from sensor glitches or manual entry

errors.

Problems Handled & Justifications:

- **Exact Duplicate Rows (750 rows):** Retaining exact duplicate records falsely inflates the statistical weight of specific observations and causes train-test contamination. We retain only the first instance using `drop_duplicates(keep='first')`.
- **Impossible Negative Values (24 rows):** Physical fabric properties cannot be negative (e.g., `tensile_strength = -12.5`, `fabric_thickness = -0.45 mm`). These represent sensor transmission dropouts. We detect them and impute them with the positive feature median.
- **Categorical Inconsistencies (120 noisy entries):** Variants like " cotton ", "COTTON", "cOtToN", and placeholder codes like "???" were normalized using `.str.strip().str.capitalize()` and mapped to valid feature categories.

5. Categorical Data Encoding

What is this process?

Machine Learning models operate on numerical matrices and gradient calculations. Categorical

encoding converts non-numeric strings into mathematical vectors.

Encoding Technique	When to Use?	Implementation Strategy	Example
Label Encoding	Ordinal categorical variables having an inherent natural rank or hierarchy.	Assign ordered integers: 0, 1, 2...	Target Variable fabric_quality: {'Low': 0, 'Medium': 1, 'High': 2}.
One-Hot Encoding	Nominal categorical variables with no inherent rank or mathematical order.	Create binary indicator columns (0 or 1) for each distinct category.	fabric_type: Cotton → [1, 0, 0, 0], Silk → [0, 1, 0, 0], Polyester → [0, 0, 1, 0], Linen → [0, 0, 0, 1].

6. Outlier Detection and Treatment

What is this process?

Outlier detection identifies observations that deviate significantly from the general distribution of

the dataset, which can adversely pull decision boundaries and distort model weights.

1. IQR Method (Interquartile Range - Non-Parametric):

- $IQR = Q_3 - Q_1$ (where Q_1 = 25th percentile, Q_3 = 75th percentile)

- Lower Threshold = $Q_1 - 1.5 \times IQR$

- Upper Threshold = $Q_3 + 1.5 \times IQR$

2. Z-Score Method (Parametric):

- $Z = (X - \mu) / \sigma$

- Potential Outlier Rule: $|Z| > 3$ (Values beyond 3 standard deviations from

the mean)

Outlier Treatment Strategy: Capping (Winsorization)

Why Cap instead of Drop? Dropping outlier rows entirely removes valid observations and reduces sample size. In industrial manufacturing, extreme fabric measurements (e.g. $gsm = 1150$) are valid high-density rolls. By **capping (clipping)** values at the upper/lower IQR bounds, we retain 100% of the dataset rows while neutralising extreme gradient instability during model training.

7. Data Transformation & Feature Scaling

What is this process?

Feature scaling transforms numerical variables onto a uniform mathematical scale so that features

with large numerical magnitudes do not dominate features with small numerical ranges.

Scaling Technique	Formula	Output Range	When to Use? (Rationale)
Min-Max Normalization	$X' = (X - X_{\min}) / (X_{\max} - X_{\min})$	[0, 1]	Ideal when the underlying data does not follow a Gaussian distribution and boundary limits [0, 1] are required (e.g., neural networks, image processing).
Standardization (Z-Score)	$Z = (X - \mu) / \sigma$	$\mu = 0, \sigma = 1$	Ideal for optimization algorithms (Logistic Regression, SVMs, PCA) as it preserves zero-centered gradients and is less sensitive to extreme ranges.

8. Train-Test Split & Data Leakage Prevention (Crucial)

What is Data Leakage?

Data Leakage occurs when information from the unseen test set is mistakenly used during data preprocessing or training, resulting in overly optimistic evaluation scores that fail in real-world production deployment.

Incorrect Leaky Workflow ❌

1. Calculate global Mean & Std across entire dataset.
2. Impute missing values and scale entire dataset.
3. Perform Feature Selection using all labels.
4. Split into Train & Test sets.

Correct Leakage-Free Workflow ✅

1. **Split raw dataset into 80% Train and 20% Test.**
2. **Fit** parameters ($\mu_{\text{train}}, \sigma_{\text{train}}$, medians) ONLY on Training data.
3. **Transform** both Train and Test data using those frozen training parameters.
4. Perform Supervised Feature Selection using Training data only.

9. Feature Selection: From Theory to Implementation

What is Feature Selection?

Feature Selection is the process of isolating the most statistically relevant subset of existing

features without changing their original physical interpretation (unlike Feature Extraction like PCA

which projects data into abstract components).

Technique	Feature Type	Target Type	Formula / Core Mathematical Logic	Decision & Real-World Justification
Variance Threshold	Numerical	Unsupervised	$\text{Var}(X) = \sum (X_i - \mu)^2 / n > \theta$	Eliminates constant or near-constant features (e.g., features with $\text{Var} < 0.1$) that provide zero discriminatory power.
Pearson Correlation	Numerical	Numerical / Linear	$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$	Identifies multicollinearity between input features. If two features have $ r > 0.85$, one can be safely removed to eliminate redundancy.
Chi-Square Test (χ^2)	Categorical	Categorical	$\chi^2 = \sum \frac{(O - E)^2}{E},$ where $E = (\text{Row} \times \text{Col}) / \text{Total}$	Evaluates statistical independence between categorical inputs (color_fastness, weave_type) and fabric quality classes.
ANOVA F-Test	Numerical	Categorical	$F = \frac{\text{Variance Between Groups}}{\text{Variance Within Groups}}$	Measures if mean values of continuous features (tensile_strength, gsm) vary significantly across quality classes.
Mutual	Discrete /	Categorical	$MI(X; Y) = H(Y) -$	Measures general non-linear

Technique	Feature Type	Target Type	Formula / Core Mathematical Logic	Decision & Real-World Justification
Information	Binned		$H(Y X) = -\sum P(x,y) \log_2[P(x,y)/(P(x)P(y))]$	statistical dependency and information gain in bits. Unlike Pearson, MI captures complex non-linear associations.

10. Final Feature Selection Decision Matrix

Feature Name	Data Type	Method Applied	Statistical Signal	Decision	Engineering Justification
tensile_strength	Numerical	ANOVA F-Test	F-Statistic = High	KEEP	Primary structural integrity metric directly determining grade durability.
gsm	Numerical	ANOVA F-Test	F-Statistic = High	KEEP	Fabric weight density is a primary commercial quality standard.
fabric_thickness	Numerical	ANOVA F-Test	F-Statistic = Medium	KEEP	Indicates weave uniformity and physical thickness specifications.
defect_count	Discrete	Mutual Information	MI = High (0.42 bits)	KEEP	Direct indicator of surface flaws and visual quality degradation.
color_fastness	Discrete	Chi-Square Test	χ^2 = High	KEEP	Standard industry dye durability test rating.
shrinkage_percent	Numerical	ANOVA F-Test	F-Statistic = Medium	KEEP	Critical measure of post-wash dimensional stability.
batch_id	Identifier	Domain	Zero Signal	REMOVE	Arbitrary metadata

Feature Name	Data Type	Method Applied	Statistical Signal	Decision	Engineering Justification
roll_number	Identifier	Inspection	Zero Signal	REMOVE	identifier that causes overfitting and memorization.
		Domain Inspection			Sequential roll counter with no physical relationship to fabric quality.
		Mutual Information			Auditor name introduces unwanted bias and zero generalization capacity.
operator_name	Categorical		MI ≈ 0.00 bits	REMOVE	
inspection_notes	Categorical	Domain Check	Target Leakage	REMOVE	Contains post-audit qualitative remarks that cause direct target data leakage.

11. Model-Based Validation (Model A vs Model B)

Validation Experiment:

We train and compare two classification models using the Decision Tree algorithm to prove the

practical impact of our feature selection pipeline.

Evaluation Parameter	Model A (All Raw/Preprocessed Features)	Model B (Selected Features Subset)	Practical Engineering Impact
Number of Input Features	23 Features	6 Core Features	74% Dimensionality Reduction

Evaluation Parameter	Model A (All Raw/Preprocessed Features)	Model B (Selected Features Subset)	Practical Engineering Impact
Training Latency (Time)	0.0482 seconds	0.0118 seconds	4x Faster Training & Real-Time Inference
Test Set Accuracy	94.80%	95.20%	Maintained & slightly improved accuracy by eliminating noise and overfitting.
Model Interpretability	Low (Complex decision tree)	High (Clean, auditable rules)	Industrial operators can inspect exact rules based on tensile strength & defects.

12. Comprehensive Viva / Demonstration Q&A (Top 5)

Q1: Why is data preprocessing required before training machine learning models?

Answer: Raw real-world datasets contain missing entries, noise, scale discrepancies, and

duplicates. Machine Learning algorithms rely on numerical optimization and mathematical

assumptions; preprocessing ensures clean, normalized inputs for stable convergence.

Q2: Why is the Median preferred over the Mean for imputing missing values in skewed

data?

Answer: The Mean is calculated as the arithmetic average and is heavily pulled by extreme

outliers. The Median is the positional 50th percentile rank and is immune to extreme values,

accurately reflecting the central distribution.

Q3: What is Data Leakage and how did your group prevent it?

Answer: Data Leakage occurs when test set knowledge influences preprocessing or training. We

strictly split our dataset into 80% Train and 20% Test splits first, computed all parameters (means,

medians, scalers, encodings) solely on the Training split, and applied those fixed parameters to

the Test split.

Q4: What is the difference between Feature Selection and Feature Extraction?

Answer: Feature Selection keeps a subset of original features without altering their physical

identity (e.g., Variance Threshold, ANOVA). Feature Extraction transforms and projects original

features into a new coordinate space (e.g., PCA, t-SNE).

Q5: Why can Mutual Information detect relationships that Pearson Correlation misses?

Answer: Pearson correlation exclusively measures linear relationships. If the relationship is non-

linear (e.g., parabolic $Y = X^2$), Pearson correlation is 0. Mutual Information measures shared

probabilistic entropy and captures all linear and non-linear relationships.